

BSCS3001

SOFTWARE ENGINEERING PROJECT MAY 2026

Milestone-1 Deliverables

Submitted by

Team Name : **Nexmind**

Team No. : **003**



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1. Project Details

Field	Value
Software Name	Campus Connect
Problem Statement	Option 1: Community Services Platform
Subdomain	Student Club Management : Higher Education Institutions
Date of Submission	28 June 2026

Team Details

Name	Roll Number	Role
Atharv Khare	23f2004201	Product Manager / Frontend Developer
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A. IDENTIFICATION OF USERS

1. Primary Users

These are the students who interact directly with the platform for their own benefit like discovering clubs, joining them, attending events, receiving results, and accessing certificates. Every workflow on the platform is ultimately designed around their experience.

Students / Club Members (e.g., Shikha Singh, undergraduate student at KNIT Sultanpur)

Represent the widest user group on the platform. Their typical activities include:

- Discovering clubs and browsing event listings
- Sending join requests and tracking approval status
- Registering for events and receiving confirmation
- Checking in at events and receiving results
- Downloading participation certificates as verifiable proof of activity
- Raising queries or issues about their registration or certificate status

Any breakdown in their experience like missed announcements, unclear join processes, lost certificates, directly reduces engagement with campus life.

2. Secondary Users

Users who log into the platform regularly to perform operational tasks. They are not end-beneficiaries of the workflow; they are the ones running it.

Club Heads / Event Coordinators (e.g., Aayansh Yadav, former head of campus gaming club; Aayush, campus ambassador; independent Claude builder club)

Responsible for creating and running clubs and their associated activities. The platform must automate the administrative steps that currently consume 7–8 hours of volunteer time per event.

- Registering a new club and tracking its approval status with the college admin
- Creating and publishing events with title, description, date, venue, and capacity
- Managing membership by reviewing and approving incoming join requests
- Marking attendance for registered participants on the day of an event
- Setting results (Winner, Runner-up, Participant) for each attendee
- Posting announcements to all club members from one place
- Reviewing and resolving member queries raised through the platform

Campus Council Admin / Student Affairs Coordinator (*e.g., Student Activity Cell coordinator, Head of Student Affairs office*)

The institutional gatekeeper on the platform. Logs in when a new club registration is submitted, less frequent than a club head but still an active platform user who takes binding decisions that determine a club's official status.

- Reviewing and approving or rejecting new club registration requests
- Configuring the college's isolated space and verified student email domain
- Monitoring overall club activity at an institutional level

3. Tertiary Users

Stakeholders and authorities who set the rules under which clubs operate. They do not participate in daily club operations but their decisions, policies, mandates, and feedback shaped what the platform needed to do, but all platform actions on their behalf are executed by the Secondary users above.

Student Body Representatives / Student Council Members

Student governance representatives whose feedback was gathered during research through a written submission. They do not use the platform directly but their concerns directly influenced the recognition and engagement design of the system.

- Provided feedback on what makes clubs credible and worth joining
- Advocated for verifiable certificates and result records as meaningful participation incentives
- Raised concerns about student accountability when joining multiple clubs without commitment
- Influenced the inclusion of the club activity leaderboard as a recognition signal

Institutional Administration (*e.g., Registrar, College Principal, Dean of Student Affairs*)

Senior institutional authorities whose policies define the boundaries within which clubs are permitted to operate. They do not interact with the platform directly. Their mandates on official recognition criteria, record-keeping requirements, and student accountability shaped the governance and compliance requirements of the system.

- Define institutional policies on which clubs may receive official recognition
- Set the standards for student participation records and certification that the platform must honour
- May consume platform-level activity reports generated for institutional review

4. Users Interviewed or Surveyed During Research:

- Shikha Singh : Undergraduate Student (Primary User)
 - Aayansh Yadav : Former Head, Campus Gaming Club (Secondary User)
 - Aayush : Campus Ambassador, Independent Builder Club (Secondary User)
 - Student Body Representative (Anonymous) : Written Feedback Submission (Tertiary User)
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B. USER RESEARCH & PAIN POINT ANALYSIS

(Based on interviews with a student, two club leaders/organizers, and a written submission from a student body representative)

1. Research Objective

To understand the real operational challenges, coordination gaps, and systemic inefficiencies that students and club leaders face when trying to participate in or manage campus clubs and events at higher education institutions.

2. Key Observational Insights

Observation 1: Everything is scattered across informal tools

Club discovery, membership, event registration, attendance, results, and certificates each happen in a different tool : Instagram DMs, WhatsApp groups, Google Forms, Google Sheets, Canva, and email, with no connection between them. Students and leaders spend more time managing tools than running actual activities.

Observation 2: Club leaders are the sole operational node

A single club leader handles every task manually: responding to DMs, filling spreadsheets, designing certificates, typing names one by one into Canva, and chasing people across multiple chat groups. When the leader is unavailable, the club stalls entirely.

Observation 3: Students have no reliable way to discover or join clubs

New students find out about clubs only through seniors or orientation. There is no centralized directory. Niche or smaller clubs have no searchable presence at all. The join process itself is informal and unconfirmed, students never know if their request went through.

Observation 4: Participation proof is fragile and unverifiable

Certificates, when issued at all, arrive weeks late, come via WhatsApp, and disappear when a student changes their phone. There is no way to verify a certificate's authenticity from outside the chat history it arrived in.

3. Stakeholder Pain Points

3.1. Students / Club Members

- No centralized place to discover clubs; learned about them only through seniors or orientation
- Joining a club required finding an Instagram page, sending a DM, filling a Google Form, waiting with no response, and then following up manually to get added to a WhatsApp group
- Missed events because announcements were forwarded only in WhatsApp groups they had not been added to
- Paid via UPI for event registration, uploaded a payment screenshot, and received no confirmation, had to message the organizer to verify the seat
- Certificates not issued at every event; when issued, they arrived 2–3 weeks late via WhatsApp and were lost when the phone was replaced
- Niche clubs (e.g., astronomy) had no online presence and were effectively invisible

Core Issue: Students have no reliable channel to discover clubs, confirm participation, or hold onto proof of their campus involvement.

3.2. Club Leaders (Gaming Club Head)

- New member onboarding was entirely manual: DMs, Sheets, WhatsApp group adds with entries frequently missed during busy periods
- Many students registered for events but did not attend; identifying actual attendees required comparing a registration spreadsheet against a handwritten attendance list after the event, with frequent errors
- Manually typing 50 participant names one by one into a Canva certificate template took 7–8 hours per event, with typos common
- Result records did not persist: "Pichle saal winner kaun tha, kisi ko pata nahi" (Nobody knows who won last year)
- Announcements posted on WhatsApp would scroll away unnoticed; the same message had to be sent multiple times
- Writing event descriptions was slow, copy-pasting from old announcements was the only shortcut
- Submitting a club proposal to the college meant going through multiple faculty approval layers with no way to track where the application stood

Core Issue: Club leaders spend most of their time on administrative repetition rather than actually running activities.

3.3. Independent Club Organizer / Campus Ambassador

- Without an official institute tag, students treated the club's messages as spam; building trust was the biggest challenge in the first weeks
- All member DMs asking about subscription or perk status had to be managed manually on WhatsApp, unsustainable at any scale
- Offline events and meetups could not be organized without official recognition, which required administrative approval that never came through in time

Core Issue: Lack of official status and structured communication channels blocks clubs from operating effectively or reaching students credibly.

3.4. Student Body Representative (Tertiary : Written Feedback)

- Certificates and recognition matter: events with no verifiable certificates or tangible outcomes see lower participation from both organizers and attendees
- Evaluating a new recruit's long-term commitment is difficult when students join multiple clubs without accountability
- Institutional inertia forces clubs into the same event formats year after year, reducing student interest over time

Core Issue: Without meaningful incentive structures and verifiable records, clubs struggle to attract and retain committed participants.

4. Cross-Stakeholder Patterns

Recurring issues across all user groups:

- Manual, disconnected tools for every step of club operations
- No confirmation or status visibility after any action (join request, registration, certificate request)
- Participation records lost or never created in a durable form
- Announcements that cannot be trusted to reach all intended recipients
- Heavy reliance on the organizer as the sole point of contact for every question

5. Root Causes Identified

- No single platform connecting club discovery, membership, events, results, and certificates
- No structured communication channel within clubs, everything routed through personal chats
- No record-keeping system for attendance, results, or certificates
- No official club registration and approval mechanism visible to all parties

- No in-platform way for members to raise queries without contacting leaders directly

6. Overall Conclusion

The demand for campus clubs and events is genuine and consistent. Students want to participate; leaders want to run activities. The breakdown is entirely operational, every step from discovering a club to holding a verifiable certificate is fragmented, informal, and dependent on the organizer's personal attention. A platform that connects these steps into a single workflow eliminates the overhead and creates the accountability that both students and institutions need.

2. USER STORIES

Separated into Epics by feature and role :

Epic 1: Platform Foundation

Story 1.1: Register a College on the Platform

- As a Campus Council Admin,
- I want to register my institution and set the official student email domain,
- So that students from my college are automatically verified and routed to their institution's space on signup.

Story 1.2: Sign Up and Log In

- As a registered user (new student or club leader) ,
 - I want to create an account with my college email and log in with one set of credentials,
 - So that I am recognized as a verified member of my institution without any manual approval step.
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Epic 2: Club Discovery and Membership

Story 2.1: Browse the Club Directory

- As a student,
- I want to browse a searchable list of all active clubs at my college with each club's name, category, and short description,
- So that I can find communities matching my interests without asking seniors or searching across social media.

Story 2.2: View a Club Profile

- As a student,
- I want to open a club's full profile page with its description, past activities, and a join option,
- So that I can decide whether the club suits me before committing.

Story 2.3: Request to Join a Club

- As a student,
- I want to send a formal join request to a club with one action on the platform,
- So that my request reaches the club leader immediately without needing to DM anyone or follow up manually.

Story 2.4: Manage Membership Join Requests

- As a club leader,
- I want to review all incoming join requests and approve or reject them from my dashboard,
- So that membership is managed in one place without me tracking individual DMs or missing bulk requests.

Story 2.5: Create a Club and Request Approval

- As a club leader,
- I want to register a new club and submit it for college admin approval through the platform,
- So that my club gains an official, trustworthy status and I can track where my application stands at any point.

Story 2.6: Approve or Reject Club registration

- As a campus council admin,
 - I want to review pending club registration requests and approve or reject each one with a written reason,
 - So that only vetted clubs receive official status and club leaders know the outcome without following up manually.
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Epic 3: Event Lifecycle

Story 3.1: Create and Publish an Event

- As a club leader,
- I want to create an event with a title, description, date, venue, and participant capacity and publish it to all members,
- So that members are informed through one action rather than me composing separate messages across WhatsApp, email, and Instagram.

Story 3.2: Register for an Event

- As a student,

- I want to register for an event and immediately receive an on-platform confirmation with a unique registration ID,
- So that I have a retrievable record that my spot is secured without needing to message the organizer.

Story 3.3: View Registered Participants

- As a club leader,
- I want to see the complete list of students who have registered for my event,
- So that I can plan logistics and use it as my baseline when marking attendance on the day.

Story 3.4: Mark Attendance at an Event

- As a club leader,
- I want to mark which registered students actually attended on the day of the event,
- So that certificates and results are issued only to genuine attendees without any post-event spreadsheet comparison.

Story 3.5: Set a Result for Each Attendee

- As a club leader,
- I want to assign a result — Winner, Runner-up, or Participant — to each student who attended,
- So that outcomes are permanently recorded and not lost in WhatsApp messages after the event ends.

Story 3.6: View My Event Result

- As a student,
- I want to see my result i.e. Winner, Runner-up, or Participant, from every event I attended, listed on my profile,
- So that I have a durable, accessible record of my participation that I can reference at any time.

Story 3.7: Edit or Cancel an Event

- As a club leader,
- I want to update event details or cancel an event if plans change,
- So that registered students always have accurate information and are not caught off-guard.

Epic 4: Communication and Issue Tracking

Story 4.1: Post a Club Announcement

- As a club leader,
- I want to post an announcement to all club members from one place,

- So that I do not have to send the same update repeatedly across WhatsApp, Instagram, and email.

Story 4.2: View Announcements from Joined Clubs

- As a student,
- I want to see a consolidated feed of announcements from all clubs I have joined,
- So that I do not miss updates regardless of which group chat or social page I happen to follow.

Story 4.3: Raise an Issue or Query

- As a student,
- I want to submit a formal issue or query about my registration, membership, or certificate directly through the platform,
- So that my concern is tracked and gets a response without me having to DM the club leader personally.

Story 4.4: View and Resolve Member Issues

- As a club leader,
 - I want to see all open issues raised by my members and mark them as resolved,
 - So that I can manage member concerns in one place rather than being flooded with individual DMs.
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Epic 5: Certificate Workflow

Story 5.1: Auto-Generate Participation Certificates

- As a club leader,
- I want the platform to automatically generate a PDF certificate for every attendee the moment I submit results,
- So that I can distribute certificates for 50 participants without opening any design tool or manually typing a single name.

Story 5.2: Download My Participation Certificate

- As a student,
- I want to access and download any of my earned certificates from my profile at any time,
- So that I have permanent proof of my participation that does not disappear when I change my phone.

Story 5.3: Verify a Certificate by Serial Number

- As a certificate holder or external verifier,
- I want to enter a certificate's serial number on a public page and confirm it is authentic,

- So that I can share verifiable proof of participation with employers or institutions without needing to log into the platform.
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Epic 6: AI-Powered Club Discovery

Story 6.1: Find Clubs by Describing My Interests

- As a student,
 - I want to describe my interests in plain language and receive a list of matching clubs with a short reason for each,
 - So that I can discover niche communities that have no social media presence and would otherwise be completely invisible to me.
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Epic 7: Recognition and Engagement

Story 7.1: Club Activity Leaderboard

- As a campus council admin,
 - I want to view a ranked leaderboard of clubs ordered by their activity score, calculated from events run, members recruited, and participation recorded,
 - So that active clubs are publicly recognised and students have a clear signal of which communities are genuinely engaged before deciding to join.
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3. FUTURE SCOPE

The following features were identified during research but are outside the scope of the current two-sprint delivery window. They are documented here to prevent re-proposal during development and to serve as a roadmap for future work.

Feature	Source	Reason Deferred
AI semantic search for clubs and events	General usability	Basic club discovery and the AI chatbot cover this in MVP; a full search engine adds infrastructure complexity. Deferred to Phase 2.
AI-assisted event description and announcement drafting	Aayansh : slow manual writing	Useful but not core to the participation spine. Requires a well-tested prompt layer. Deferred to Phase 2.
AI event recap generator	Aayansh : no post-event summary process	Dependent on structured result data being stable first. Deferred to Phase 2.
Member performance tracking for recruitment decisions	Student body feedback	Privacy-sensitive design challenge. Needs careful scoping before implementation.
Cross-college or federated club discovery	General scalability	Multi-tenant federation requires architecture decisions beyond current scope.
Event expense and budget tracking	Student body feedback (incentives)	No interviewee cited this as a primary pain in the MVP context. Deferred to Phase 2.

4. INTERVIEW AND OBSERVATION EVIDENCE

Proof of user research as required by the course rubric:

- **Interview 1** : Shikha Singh (Primary User, Undergraduate Student, KNIT Sultanpur):
VIDEO LINK -
https://drive.google.com/file/d/1mUlGiJAR9Fwf6j5U4fNYShJWl4g_4Ax_/view?usp=sharing
- **Interview 2** : Aayansh Yadav (Secondary User, Former Head, Campus Gaming Club):
VIDEO LINK -
https://drive.google.com/file/d/1XouDKw3aIQSxIttzvc_w_w8oereGNY1/view?usp=drive_link
- **Interview 3** : Aayush (Secondary User, Campus Ambassador, Claude Builder Club):
VIDEO LINK -
https://drive.google.com/file/d/1gGb2VxNalnMeyM19ZnltofngSxD1QxD2/view?usp=drive_link
- **Observation** : Student Body Representative Written Feedback (Tertiary User):
DOCUMENT LINK -
https://drive.google.com/file/d/1I8TfOsKgYi_RA290wkEXitVhaH0f45Wh/view?usp=drive_link